



**Politecnico
di Torino**

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Software Design and Architecture

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Capitolo 1

Introduction to Software Design

1.1 Complexity

Complexity is anything related to the structure of a software system that makes it hard to understand and modify the system.

$$C = \sum c_p t_p$$

Where C is determined by all software parts p , c_p is the complexity of part p , and t_p is the fraction of time spent on part p .

1.1.1 Syntax complexity

- Change amplification: change require operating in different places
- Cognitive load: how much information to learn to understand and perform a task. Some times is better to write longer code to reduce the complexity and so the cognitive load.
- Unknown unknowns: there are partes of the system that we don't know where they are or even that they exist.

1.1.2 Cause of complexity

Dependency: When we need to understand or modify a part of the code, we need to understand or modify other parts of the code. Try to reduce the number of dependencies and the complexity of the dependencies.

Obscuiry: alla the important information shoul be clearly visible.

1.1.3 Strategical vs. Tactical Programming

- **Tactical programming:** focus on having code that works as fast as possible. "Move fast and break things".
- **Strategical programming:** focus on long term maintainability and scalability. "Working code is not enough".

1.2 Modules

To improve and eliminate dependency and complexity, we can use modules. A module is a part of the code that has a well defined interface and a well defined implementation.

- **Interface:** what we need to know to use the module. Formal part, possible checked by the compiler. Represents a cost.

- **Implementation:** how the module works. Ideally much more complex than the interface. Represents a benefit.

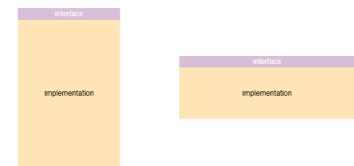
1.2.1 Abstraction

What we typically do is to perform an abstraction. Abstraction means selective removal of information. Is a simplified view of the something that omits "unimportant" information. Omitting information can lead to errors and keeping unimportant information can lead to cognitive overload.

Deep modules and shallow interfaces

Deep modules: modules with a simple interface and a complex implementation. They are good because they reduce the cognitive load and the change amplification. They are bad because they can lead to unknown unknowns.

Shallow interfaces: modules with a complex interface and a simple implementation. They are good because they reduce the unknown unknowns. They are bad because they can lead to cognitive overload and change amplification.



1.2.2 Information Hiding

Information hiding is the principle of hiding the implementation details of a module from the users of the module. Example sort function: we want to sort a list of numbers, we don't care about how the sorting algorithm used.

1.2.3 Information Leakage

Is the opposite of information hiding. Occurs when the implementation details of a module is reflected in multiple modules with consequences of creating dependencies.

1.2.4 Information Hiding Red Flags

- **temporal decomposition:** execution order is reflected in the code structure. Operations that happen in different times are in different classes.
- **overexposure:** API forces knowledge about rarely used features to use common ones. E.g.
`BufferedReader r = new BufferedReader(new InputStreamReader(System.in));`
- **Too many classes:** code is divided into too many shallow classes.

1.2.5 Method separation

Better together or Better apart?

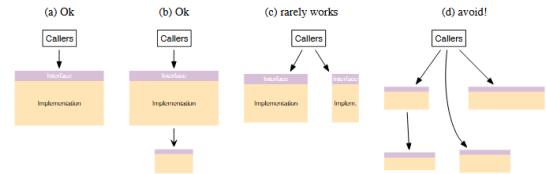
- **Bring together:** if information is shared, if we simplify the interface, if we eliminate duplication.
- **Separate:** if the code is general purpose or specific purpose.

Method separation red flags

If we have a lot of repeated code and/or we mixed specific and general code, means we have put together code that should be separated.

1.2.6 Do one simple thing and complexity

Each method should do one thing and do it completely



1.3 Obscurity

Obscurity is the principle of making the important information clearly visible and the unimportant information hidden. Obscurity is a great source of complexity. Obscure code is hard to Understand and creates more Bugs.

The solution to obscurity is writing obvious code, this means with a quick glance we can understand what the code does. We can achieve this by using abstraction layers.

1.3.1 Abstraction layers

Software systems are composed in layers. Higher layers use the facilities provided by lower layers.

Abstraction simplifies understanding and reduces obscurity.

In a well-designed system, each layer provides a different abstraction from the layers above and below it.

Abstraction Layers Red Flags

Some red flags that indicate that we have a problem with abstraction layers are:

- **Pass-Through methods:** is a method that does nothing but call another method.
- **Pass-Through variables:** is a variable that is passed through over and over down a long chain of method calls.

1.3.2 Writing Comments

Another method to reduce obscurity is writing comments. If users need to read a method code to use it, there is no abstraction.

Comments should describe things that are not obvious in the code:

- **Pick conventions:** write comments like other persons in the project
- **Don't repeat code:** avoid redundant comments that simply restate what the code already makes clear
- **Higher level comments enhance intuition**
- **Delayed comments are bad comments**
- **Use comments as design tool**

1.3.3 Naming

Toggether with comments, naming is a great tool to reduce obscurity. Bad names cause bugs. Name should create a image in mind of the reader about the nature of the thing being named. Names should be precise and consistent.

1.4 Errors and Anomalies

Exceptions add complexity because they create dependencies between the code and if they are easy to throw they are hard to handle.

A piece of code might throw exceptions in different ways:

- Caller may provide bad arguments or configuration
- Invoked method may not be able to complete the request
- In distributed systems network packets may be lost or delayed or peers may communicate in unexpected ways
- The code may detect bugs or internal inconsistencies or situations that are not prepared to handle

1.4.1 Define Errors out of Existence

What are the solutions to errors? Define errors out of existence. This means, not ignore the error but to redefine the behavior of a method so that no error is required.

E.g. `String.substring(int beginIndex, int endIndex)` it throws different exceptions in only specific cases. This is completely avoidable and unnecessary.

1.4.2 Limit exceptions

There are some methods to limit the number of exceptions that a method can throw:

- Handle exceptions at the point where they are thrown, if possible
- Exception aggregation means handle different exceptions with same handler
- Just crash
- Mask exceptions avoiding unexpected exceptions

1.5 Dependency Dimensions

1. Structural (Code-Level) Dependencies
2. Runtime / Distribution Dependencies
3. Data-Level Dependencies
4. Behavioral Dependencies

1.5.1 Structural (Code-Level) Dependencies

- **Implementation Dependency:** some piece of code depends on a concrete class instead of an abstraction
- **Construction Dependency:** you depend on how a collaboration, another class, is created.
- **Compile-Time Dependency:** the module knows something at compile time

1.5.2 Runtime / Distribution Dependencies

Often linked to distributed architectures.

- **Spatial Dependency:** A component depends on the location of another component. E.g. hard-coded URLs
- **Temporal Dependency:** A component depends on another being available at the same time. E.g. remote method calls or servers
- **Synchronization Dependency:** A component depends on another to complete work before continuing. E.g. synchronous request/response

1.5.3 Data-Level Dependencies

- **Data Dependency:** One component depends on another's internal data model. E.g. shared database
- **Schema Dependency:** Multiple services depend on the same database schema or representation structure. E.g. DB schema, XML/Json schema

1.5.4 Behavioral Dependencies

- **Control Flow Dependency:** One component dictates another's execution path. E.g. Centralized process managers
- **Transactional Dependency:** E.g. Components rely on atomic distributed consistency.
- **Timing Assumption Dependency:** Assumptions about response time or ordering. E.g. "Service must respond in <50ms" or Event ordering assumptions

Capitolo 2

Design Patterns

Initially proposed by Christopher Alexander in the context of architecture.

A reusable **solution**
to a known **problem**
in a well defined **context**

The context is a *design* situation giving rise to a (design) problem.

The problem is a set of forces repeatedly arising in the context.¹

The solution is a proven resolution of the problem. Configuration to balance forces. Structure with components and relationships and run-time behaviour.

Example of Social Pattern: You are in queue to the supermarket gastronomic section. The problem is hard to spot who arrived first. The solution is to use a ticket machine.

2.1 Types of Software Patterns

- **Architectural patterns** - high level structures of software systems. E.g., layered architecture, client-server, microservices.
- **Design patterns** - solutions to common design problems. E.g., Singleton, Observer, Factory Method.
- **Idioms** - language-specific patterns. E.g., RAII in C++, list comprehensions in Python.

2.1.1 Architectural pattern

Normally the architectural pattern works with big components and their interactions and is needed to describe the responsibilities of these components. Expresses a fundamental structural organization schema for software systems and defines the rules and guidelines for organizing the relationships between the components.

Example of architectural pattern: Several programs that are used in sequence read from input and write sequentially to output.

The problem is there are a lot of intermediate files used for communication between programs.

The solution is to adopt pipes and filters architecture feeding with the result of the previous one.

2.1.2 Design patterns

Provides a scheme for refining components of a software system or their relationships and describes a commonly recurring structure of communicating components.

¹Any relevant aspect of the problem E.g., requirements, constraints, desirable properties

Example of design pattern: A class library providing few functionalities contains a lot of classes. The problem is the user is exposed to the internal complexity of the library. The solution is to create a new façade class that interacts with the user and hide all the details.

2.1.3 Idioms

It is a low-level pattern specific to a programming language. Describes how to implement particular aspects of components or the relationships between them. Leverages the features of a programming language.

Example of idioms: An attribute is constant and should be globally available to many classes. The problem is opening access would allow unauthorized modifications and the attribute is repeated in every object. The solution is to make public static final in Java language.

2.2 Pattern language

Pattern do not exist in isolation in fact can happen that two or more patterns are applied together, used to implement part of another pattern and introduce a problem solved by another

They are called Pattern Languages or pattern systems.

Collection of patterns together with guidelines for: Should:

- Implementation
- Combination
- Practical use
- Count enough patterns
- Describe patterns uniformly
- Present relationships

2.3 Idioms

X-ability

Clonability, serializability, comparability, etc. some specific behaviour dealing with many heterogeneous classes. The solution is to derive from a class or interface that defines the required methods.

Example: In Java all methods extends the root class *Object* and provides the methods *toString()*, *equals()*, *hashCode()*, etc. that can be overridden to provide specific behaviour.

2.3.1 Strong typing vs. Duck Typing

"If it looks like a duck, swims like a duck, and quacks like a duck, then it probably is a duck."

Strong typed language: You must define interfaces and classes to be able to call a specific method. Compile time type checking.

Dynamic/duck typed language: You just add a method to a class and you can call it. No compile time type checking.

2.3.2 Property

A property, more general than an attribute, is a named characteristic of an object typically accessed through well-defined operations called accessors, getters, and setters forming the public interface of the object and it is not necessarily an attribute of the object (e.g. the age can be derived from the date of birth).

2.4 Design Patterns

Description of communicating objects and classes that are customized to solve a general design problem in a particular context.

A design pattern names, abstracts, and identifies the key aspects of a common design structure that make it useful for creating a reusable object- oriented design.

		Purpose		
		Creational	Structural	Behavioral
Scope	Class	1	1	2
	Object	4	6	10

2.4.1 Pattern selection

How to use patterns? First of all you need to know the problem you are trying to solve, knowing the patterns and then select the one that best fits your needs.

2.4.2 Creational patterns

- Class

Factory Method

- Object

Abstract Factory

Builder

Prototype

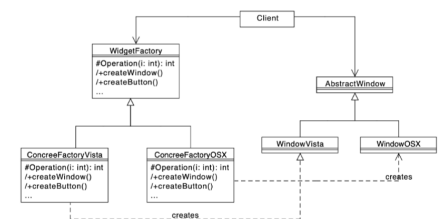
Singleton

Abstract Factory

A family of related classes can have different implementation details.

The client should not know anything about which variant they are using / creating.

Example: GUI where you have different components (buttons, text fields, etc.) and the same GUI can be applied to different platforms such as Windows, MacOS, Linux, etc.

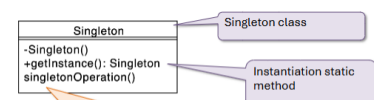


Singleton

A class represents a concept that requires a single instance.

The client should be able to access, in appropriate way, the unique instance.

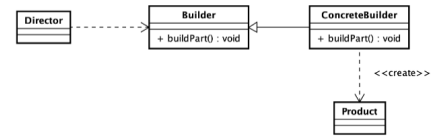
Note: If we talk about of an object designer pattern, in python you can just create a global variable and it will be a singleton.



```
private Singleton() { }
private static Singleton instance;
public static Singleton getInstance(){
    if(instance==null)
        instance = new Singleton();
    return instance;
}
```

Builder

An object of a complex class has to be created.
The creation entails complex interaction with the object and could be different variations of the object.



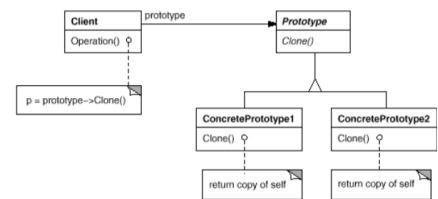
Example: $10.40 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-3}$

```
//Usual non-fluent builder
Measure power = new Measure(10.4);
power.addUnit("kg", 1);
power.addUnit("m", 2);
power.addUnit("s", -3);
power.setPrecision(2);
```

```
//Fluent builder
Measure power = new Measure.Builder(10.4)
    .is("kg")
    .by("m")
    .squared()
    .by("s")
    .to(-3)
    .withPrecision(2)
    .build();
```

Prototype

Several object can be created out of a palette of similar classes.
The class hierarchy can become quite complex and classes in hierarchy are differentiated by minor parameters (that could be defined through constructor arguments).
A solution could be start with a prototype classe and clone it to create new objects.



Example: A graphical editor where you have very similar shapes (e.g. square with different radius, colors, etc.).

2.4.3 Structural patterns

How organize classes and objects to compose larger structures.

- Class

Adapter

- Object

Bridge

Composite

Decorator

Facade

Flyweight

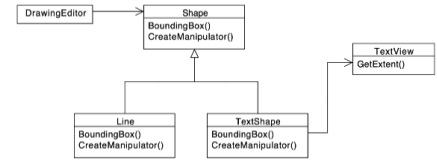
Proxy

Adapter

A class provides the features required by another class but its interface is not the one expected by the client.

The integration of the provider class should be possible without modifying it. The source code could be not available or is already used as it is somewhere else.

Example: In Java, keyboard events are handled by `KeyListener`. Listener class need to implement the interface and implement all the methods. If you are interested only in one of the methods, you can use the `KeyAdapter` class that is empty and you can decide which method to override, and the Listener extends `KeyAdapter` instead of `KeyListener`.

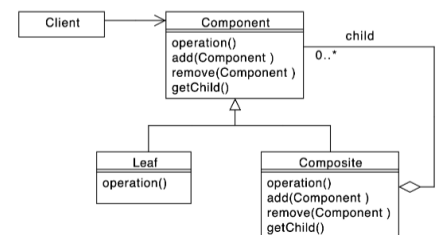


Composite

You need to represent part-whole hierarchies of objects.

Clients can be very complex and they require to adapt the difference between all the elements in the hierarchy.

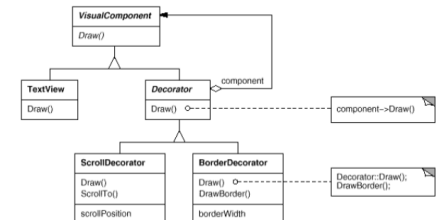
Example: In Java UI components are organized in a tree structure. You can have a panel that contains other panels and buttons, etc.



Decorator

You have some objects of classes that have specific behaviour and you want to add the same behaviour plus something else dynamically.

The solution can be extend the class but if that behaviour is something you would to apply to many classes, you need to extend all of leaf classes. Using a decorator you can modify the behaviour without creating an explosion of subclasses for every combination of features

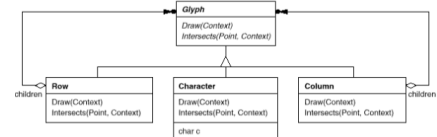


Example: In Java we have `JScrollPane` can be used to provide a scroll pane to any component

Flyweight

You need need to create a very large number of similar objects, and memory usage and performance become concerns.

Example: In Java we can pool different objects together. The `Integer` class in Java.



Proxy

You need to controll access to an object.

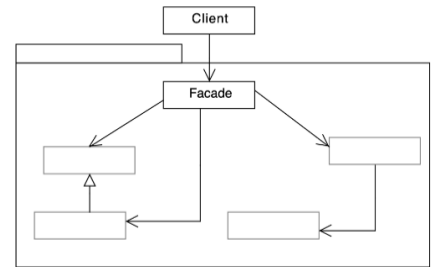
We need to provide a placeholder that is able to mediate between the client needs and all the requirements you want to controll before accessing the final object without changing interface or class.

- **Remote proxy:** provides a local representative for an object in a different address space.
- **Virtual proxy:** creates expensive objects on demand. E.g. slow loading resource - images in a web page.

- **Protection proxy:** controls access to the original object. E.g. when objects should have different access rights.
- **Smart reference:** is a replacement for a bare pointer that performs additional actions when an object is accessed. E.g. counting the number of references, checking permissions, etc.

Facade

Hide the details and the complexity of a general class by providing a simpler interface. The client should not know anything about the subsystem.



2.4.4 Behavioral patterns

Behavioral patterns allow to implement specific algorithms using objects and classes to provide control and flexibility.

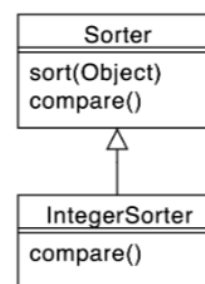
- Object
 - Chain of Responsibility
 - Command
 - Iterator
 - Mediator
 - Memento
 - Observer
 - State
 - Strategy
 - Visitor
- Class
 - Interpreter
 - Template Method

The basic mechanism of behavioral patterns is encapsulation variation. Instead of having a lot of if-else you have a structure at object level that allows you to define different alternatives.

2.4.5 Template Method

An algorithm/behavior has a stable core and several variation at given points. You have to implement/maintain several almost identical pieces of code.

Example: The compare in Java. You have to implement the compare method that solves integer or string comparison.



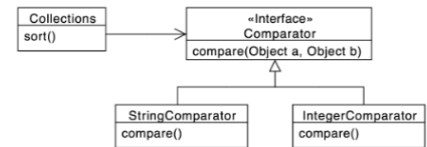
2.4.6 Strategy Pattern

Many classes or algorithm has a stable core and several behavioral variations, several different implementations are needed. Every variation of an algorithm can be embedded in an object and passed as a parameter to the algorithm.

Example: Comparator in Java.

Consequences:

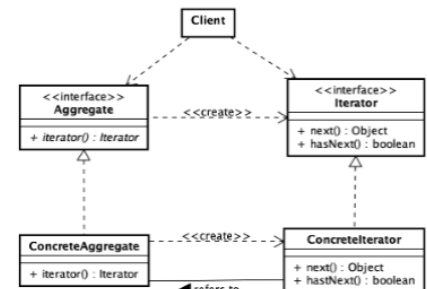
- + Avoid conditional statements
- + Algorithms may be organized in families
 - + Choice of implementations
 - + Run-time binding
- Clients must be aware of different strategies
 - Communication overhead
 - Increased number of objects



2.4.7 Iterator pattern

You have a collection of objects and you want to iterate on it allowing multiple concurrent iterators.

Example: In Java there is the Iterable interface and Iterator.



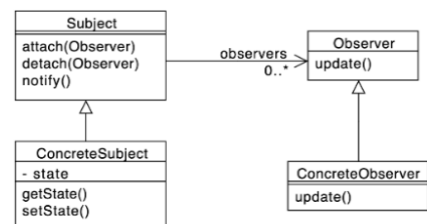
2.4.8 Observer pattern

The change in one object may influence one or more other objects. This can cause high coupling between objects and the number and type of objects to be notified may not be known in advance.

Example: In Java there is the Observable interface and Observer interface. This is very common in UI programming.

Consequences:

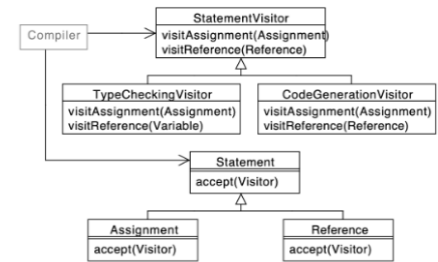
- + Abstract coupling between Subject and Observer
- + Support for broadcast communication
 - Unanticipated updates



2.4.9 Visitor

You have an object structure that contains many classes with different interfaces and you want to perform several different and unrelated operations on these objects.

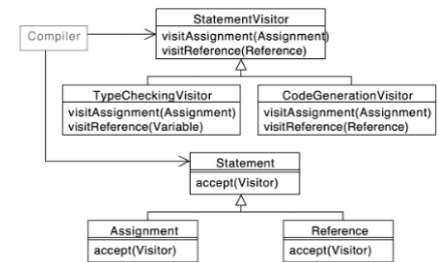
The operations depend on the concrete classes and you want to avoid putting many different methods in each of these classes.



2.5 Inheritance vs. composition

Reuse can be achieved via:

- **Inheritance:** a new class is created by inheriting from an existing class and adding new features or modifying existing ones. Clients can invoke directly inherited methods.
- **Composition:** The class has a link or instance to a class you want to reuse and just forwards the calls to it.



Note: The Observable pattern is deprecated. The PropertyChangeSupport is more flexible.

2.6 Abstract Syntax Tree

The AST is a tree that represent a syntax of formal language. We can use the composite pattern to representation the arithmetic expressions.

And using the visitor pattern or the interpreter pattern to evaluate the expression.

2.7 Model View Controller